Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/05/2015
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Project Title:	Spatiotemporal Evolution of Three-Dimensional Dendritic Array Structures (SPADES)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	06/12/2012	**End Date:**	06/11/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 6/11/2016 per L. Strutzenberg/MSFC and NSSC (Ed., 5/6/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Karma, Alain (Northeastern University)		
Grant/Contract No.:	NNX12AK54G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Cellular and dendritic microstructures formed during industrial solidification processes (such as casting, welding, and brazing) determine the properties of commercial metallic objects produced in numerous industries, with applications ranging from automobile engine parts to aircraft turbine blades. From a fundamental standpoint, the formation of these structures is controlled by a subtle interplay between microscopic interfacial phenomena, and macroscopic transport of heat and solute, which is strongly influenced by gravity-induced melt convection under terrestrial conditions. Therefore, microgravity directional Bridgman experiments provide a unique means to investigate microstructure development with negligible convection and to test the predictions of theoretical models in a purely diffusive growth regime. While extensive experimental and theoretical work during the past several decades has yielded basic insights into complex solidification patterns, their dynamics in three dimensions (3D) still remains poorly characterized and understood. Fluid convection on Earth leads to large scale inhomogeneities of temperature and alloy concentration in bulk samples. Those inhomogeneities make it difficult to probe the spatially extended pattern dynamics under well-controlled diffusive growth conditions with constant growth velocity, temperature gradient, and alloy composition. For this reason, basic experimental studies of interface dynamics have been restricted to thin samples where fluid flow is eliminated. In addition, computational modeling of 3D patterns has remained challenging due to the several orders of magnitude disparity between capillary and transport scales. Detailed experimental studies in a microgravity environment where (i) convection is reduced sufficiently to obtain growth under diffusive conditions and (ii) the interface evolution can be directly visualized in three dimensions provide an unprecedented opportunity to elucidate fundamental mechanisms of the formation of those hierarchical structures by providing quantitative benchmark data that can be compared to predictions of computational models. The goal of this work is therefore to (i) obtain extensive benchmark experimental data under microgravity conditions over a range of velocity, thermal gradient, and alloy composition in a transparent system by imaging the dynamic evolution of microstructures under directional solidification conditions in the Directional Solidification Insert (DSI) of the DECLIC facility (DEvice for the study of Critical LIquids and Crystallization) installed onboard the International Space Station (ISS), and (ii) use state-of-the-art phase-field simulations to make quantitative predictions of dendritic hierarchical structure formation under diffusive growth conditions that can be directly compared to experimental observations under prescribed growth conditions. We expect the combination of those experimental and computational studies to establish a reliable framework for predicting dendritic microstructures in a wide range of alloy systems of technological relevance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based superalloys used in turbine blades to light weight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. By bringing together well-controlled and accurately measured three-dimensional solidification experiments with quantitative phase-field simulations, this research should bring significant progress in understanding the origin of side-branching and the nature of the cell to dendrite transition, which have been a long standing issue in solidification for several decades. The deeper understanding of microstructure formation dynamics shall thus accelerate and optimize the design of new technological alloys and their processing conditions that can directly benefit US industries.
Task Progress:	In the cellular regime, the present microgravity experiments reveal a rich variety of 3D breathing modes with limited spatiotemporal coherence. At low growth velocities and high enough temperature gradient, cellular arrays exhibit a global oscillatory behavior with an oscillation period homogeneous over the spatially extended array. However, phase locking does not occur throughout the whole system, but only very locally among neighbor cells. Such behaviors were reproduced with phase-field simulations with a remarkable accuracy on the predicted oscillation period (e.g., experiments: t ˜ 45.6 min, phase-field: t ˜ 44.6 min, at G = 28K/cm and V = 1µm/s). Moreover, simulations clarified the underlying physics of this oscillatory behavior: (i) the oscillatory behavior occurs at high temperature gradients at the end of a stable spacing branch when a gap in the stable array spacing appears; (ii) the lack of global coherence in experiments is due to the spatial reorganization of cells, maintained notably by tip splitting events; (iii) breathing mode oscillations are possible over a narrow range of growth velocities for control parameters that does not lead to gap in the stable spacing branch. The identification of this subtle oscillatory mechanism constitutes the first quantitative agreement between observed and calculated dynamics in a three-dimensional configuration. In the dendritic regime, experiments on a SCN - 0.24 wt% camphor alloy suggested the existence of possible coherent modes of dendritic sidebranching for high growth velocities. This mechanism of coherent side branching was earlier reported in thin sample experiments on a dilute SCN alloy as well as in 2D and thin-sample 3D phase-field simulations, showing that this mechanism of coherent branching events is related to a limit cycle. We confirmed the existence of this coherent branching mechanism with full 3D phase-field simulations. We are currently investigating the link between the spatial order of spatially extended dendritic arrays and their dynamics of growth and sidebranching. Microstructures in directional solidifications are a consequence of their thermal history from initial planar interface dynamics to coarsening or eliminating dynamics between cells/dendrites. Thus, we have investigated the initial planar solid-liquid interface dynamics with experiments on Earth and onboard the ISS. The experiments on Earth show the presence of solute boundary layers and a slow motion of the solid/liquid interface within a fixed temperature gradient before the sample is pulled. Calculations with a sharp interface model confirm this initial interface movement caused by diffusions in bulk phases. Experiments were completed on the effect of initial boundary layer on the subsequent planar front dynamics and a numerical model was developed to characterise the initial boundary layer from the experimental measurements of interface temperature and the gradient of interface temperature with time when the motion of the sample was initiated to start the directional solidification process. The initial interface dynamics in directional solidification has also been studied based on microgravity experiments data. We identified an initial drift of the thermal profile in the crucible, which affects the initial planar front dynamics. We reproduced experimental measurements by modifying the classical Warren-Langer model to account for this thermal drift. Even though simulations provide quantitative predictions on oscillatory modes at low velocity and coherent sidebranching at high velocity as in microgravity experiments, average cell spacing in simulations are lower than experiments onboard the ISS. In order to study spacing selection mechanism, we have introduced a thermal drift into the PF, which leads to a higher cell spacing, hence closer to the experimental measurements. Additionally, we addressed the effect of crystal orientation in spacing selection. The experimental observations show that, after morphological destabilization and initial coarsening dynamics, cells in a grain are generated in an upstream region, move with an angle, and are eliminated near sub grain boundaries, ultimately leading to spacing inhomogeneities. Related PF simulations

	confirm that the crystal orientation affects spacing selection of a microstructure. Therefore, we are currently investigating microstructure selection under the influence of both thermal drift and crystal orientation. In the DECLIC-DSI experiments, dendrites were only observed in a high velocity regime where the solidification front exhibited significant curvature due to the combination of the rejection of latent heat and the much lower thermal conductivity of the alloy than its quartz container wall. In addition, dendrite tip radii are too small to be accurately resolved at those high velocities. We expect the DSI-R experiments with a higher camphor composition (SCN - 0.5 wt% camphor) to address these limitations by reaching dendritic structures at more moderate velocities and with a higher tip radius that we could directly measure by interferometry analysis. Furthermore, uncertainties remain with respect to the accurate identification of materials and control parameters. In order to limit those uncertainties, we performed an extensive study of the SCN - 0.5 wt% camphor alloy using both ground-based thin-samples experiments, and quantitative 3D phase-field simulations. We confirmed that well-developed dendritic array structure form at a lower growth velocity, which will limit the curvature of the solidification front in future microgravity experiments. Also, measured and simulated dendrite tip radius showed very good quantitative agreement and a higher radius, which could be directly and accurately measured by interferometry. We have also undertaken a study on the critical influence of sample thickness on microstructure selection in Earth-based thin sample experiments. Simulations and ground-based experiments are in progress to develop the flight matrix of growth conditions. In addition, the implementation of thermocouples in the device would be extremely useful for reliable and direct measurement of the thermal conditions and would represent a precious information thus limiting uncertainties about control parameters in future microgravity experiments. Ground-based experiments were also performed to determine the phase diagram of succinonitrile-camphor system, specifically to determine the solidus line since two different solidus lines have been presented in the literature.
Bibliography Type:	Description: (Last Updated: 04/03/2024)
Articles in Peer-reviewed Journals	Xu M, Fabietti LM, Song Y, Tourret D, Karma A, Trivedi R. "Initial dynamics of a solid-liquid interface within a thermal gradient." Scripta Materialia. 2014 Oct 1;88:29-32. http://dx.doi.org/10.1016/j.scriptamat.2014.06.006 , Oct-2014
Articles in Peer-reviewed Journals	Fabietti LM, Mazumder P, Trivedi R. "Dynamics of planar interface growth during directional solidification of alloys." Scripta Materialia. 2015 Mar 1;97:29-32. http://dx.doi.org/10.1016/j.scriptamat.2014.10.019 , Mar-2015
Articles in Peer-reviewed Journals	Mazumder P, Fabietti LM, Trivedi R. "Numerical model of planar interface dynamics during directional solidification of alloys, Supplementary Material." Scripta Materialia. 2015 Mar;97:29-32. , Mar-2015
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Awards	Bergeon N, Tourret D, Chen L, Debierre J-M, Guerin R, Ramirez A, Billia B, Karma A, Trivedi R. "One of the top discoveries in microgravity on International Space Station, Awarded by the American Astronautical Society, May 2014." May-2014
Awards	Trivedi R. "John Hunt Medal for outstanding contributions to the science and technology of solidification from the Institute of Materials, London. June 2014." Jun-2014